

Equivalence Partitioning, Boundary-Value Analysis, and Negative Testing

SE 413/513 — Software Engineering

Course Reading Material

Unit: Systematic Test-Case Design Techniques

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```
@misc{hasan2026stesting,  
author = {Hasan, Kamrul},  
year = {2026},  
title = {Software Testing},  
institution = {Grand Valley State University},  
url = {https://github.com/mdkamrulhasan/software-testing}  
}
```

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1 Learning Objectives

After completing this reading, you should be able to:

- Explain why systematic test-case design techniques are necessary, given that exhaustive testing is impossible.
- Identify equivalence classes (valid and invalid) for a given input specification.
- Apply equivalence partitioning to select a small, representative set of test inputs.
- Explain why defects cluster at boundaries, and apply boundary-value analysis to target them.
- Combine equivalence partitioning and boundary-value analysis into one coherent test design.
- Define negative testing and distinguish it from positive testing.
- Design negative test cases that check for graceful failure rather than crashes or silently wrong output.
- Translate a systematic test design into a parametrized `pytest` suite.

2 Why We Need Systematic Test-Case Design

Recall Principle 2 from the *Testing Foundations* reading: **exhaustive testing is impossible**.^[1, 5] Even a function with a single 32-bit integer parameter has over four billion possible inputs; a function with two such parameters has over 1.8×10^{19} . No team can run, read, or maintain that many test cases — and running them would not even help, since Principle 1 reminds us that passing tests only shows the *absence* of the failures we specifically checked for.

This raises a practical question that this reading answers directly: **if we cannot test every input, which inputs should we test?** Picking inputs at random, or just picking “some easy-looking numbers,” tends to produce test suites that are both bloated (many redundant cases that all exercise the same code path) and full of holes (whole categories of input, especially unusual or invalid ones, never tested at all).

Equivalence partitioning and **boundary-value analysis** are two fundamental specification-based techniques for choosing test inputs systematically.^[2, 5, 4] Together with **negative testing** — the deliberate practice of testing invalid input — they form the core toolkit for designing a test suite that is both small and effective.

Key Idea

These are **black-box** techniques: they are derived entirely from what the software is *supposed* to do (its specification or requirements), not from reading its source code.^[2] Contrast this with **white-box** techniques (such as statement or branch coverage), which are derived by looking at the code’s internal structure. Black-box design is what you use before code even exists; white-box coverage is what you use afterward, to check whether your black-box tests actually exercised the implementation.

3 Equivalence Partitioning

3.1 The Basic Idea

Equivalence partitioning (EP) divides the space of possible inputs into a small number of **equivalence classes** (also called equivalence partitions),^[2, 4] such that the software is expected to behave the *same way* for every input within a given class. The core assumption is: *if the software works correctly for one representative value in a class, it probably works correctly for all values in that class*.

Instead of testing every possible input, you test **one representative value from each class**. This turns an infeasible testing problem into a small, manageable one.

3.2 Valid and Invalid Classes

Every input specification implicitly (or explicitly) divides inputs into:

- **Valid equivalence classes:** inputs the software is specifically designed to accept and process normally.
- **Invalid equivalence classes:** inputs that fall outside the specification — out of range, wrong type, malformed — for which the software should still behave in some well-defined way (typically, reject the input cleanly).

Common Pitfall

It is easy to design test cases only for valid classes and forget invalid ones entirely. A test suite with excellent coverage of valid inputs and zero coverage of invalid inputs tells you nothing about how your software behaves when it inevitably receives bad data — and bad data is inevitable. We return to this in Section 5.

3.3 Worked Example: Username Validation

Recall `is_valid_username(username)` from the *pytest* reading's Exercise 1, specified to return `True` for usernames between 3 and 20 characters (inclusive), and `False` otherwise. Applying equivalence partitioning to the *length* of the input gives three classes:

Equivalence Class	Description	Example Value
EC1 (invalid)	$\text{Length} < 3$	"ab" (length 2)
EC2 (valid)	$3 \leq \text{Length} \leq 20$	"alice" (length 5)
EC3 (invalid)	$\text{Length} > 20$	a 25-character string

Instead of testing dozens of different lengths, three well-chosen test cases — one per class — give you confidence about the entire input space, under the equivalence assumption. Of course, this assumption can be wrong (perhaps the implementation has a special, undocumented case for length exactly 10); boundary-value analysis, covered next, specifically targets the place where that assumption is most likely to break down.

3.4 Multiple Parameters

Real functions often take more than one parameter, and each parameter may have its own equivalence classes.[2, 4] If a function has two parameters with 3 classes each, testing every *combination* requires $3 \times 3 = 9$ test cases; three parameters with 3 classes each requires 27; the combinations grow multiplicatively. In practice, teams typically test each class individually (holding other parameters at a typical valid value) rather than every combination, and reserve full combinatorial or pairwise testing — a more advanced technique for when parameter *interactions* are a specific risk — for cases where the risk of combination-specific defects is high enough to justify the added cost.

4 Boundary-Value Analysis

4.1 Why Boundaries Matter

Testing guidance and established test-design literature emphasize that behavior at partition boundaries is more likely to be incorrect than behavior within the interior.[2, 5] This should not be surprising — boundaries are exactly where a programmer is most likely to make an off-by-one mistake: writing `<` instead of `<=`, `>` instead of `>=`, or getting confused about whether a range is inclusive or exclusive at one or both ends.

Key Idea

Boundary-value analysis (BVA) is the technique of deliberately testing *at* and *immediately around* the edges of each equivalence class,[2, 5] rather than relying on one arbitrary representative value from the middle of the class. It is a direct, evidence-based refinement of equivalence partitioning: EP tells you *which classes* exist; BVA tells you *which values within (and around) those classes* are most worth testing.

4.2 The Technique

For each boundary in the specification, BVA tests three values: the boundary value itself, the value immediately below it, and the value immediately above it. For a class specified as $3 \leq \text{length} \leq 20$, the boundary at 3 is tested with 2, 3, and 4; the boundary at 20 is tested with 19, 20, and 21.

Boundary	Just Below	At Boundary	Just Above
Lower bound (3)	2 (invalid)	3 (valid)	4 (valid)
Upper bound (20)	19 (valid)	20 (valid)	21 (invalid)

4.3 A Boundary Bug in `calculate_late_fee()`

Recall the fault discussed in the *Testing Foundations* reading:

```
def calculate_late_fee(days_late, daily_rate=1.50, max_fee=25.00):
    fee = days_late * daily_rate
    if fee > max_fee:
        return max_fee
    return fee
```

The condition `fee > max_fee` has a boundary at `fee == max_fee` exactly. Applying BVA here (rather than only testing an arbitrary “large” value like 30 days) means specifically testing the input that makes `fee` equal to `max_fee`:

```
>>> calculate_late_fee(days_late=25/1.50)    # fee == 25.0 exactly
25.0
```

In this particular case, the boundary happens to be handled correctly (`fee > max_fee` is false when they are equal, so the function returns `fee` itself, which equals `max_fee` anyway). But had the developer written `fee >= max_fee` instead, or capped using `min(fee, max_fee)` incorrectly, this exact boundary value is precisely where the mistake would surface — and an arbitrary “large” test value like `days_late=30` would never expose it, because it lands safely inside the interior of the capped region.

Tip

A useful habit: whenever you see a comparison operator (`<`, `<=`, `>`, `>=`, `==`) in a specification or in code you are testing, ask “what is the exact value where this condition flips from true to false?” — and write a test for that exact value.

Real-World Case Study

The Zune 30GB Freeze (December 31, 2008).[6, 7] On the last day of 2008 — a leap year — every Microsoft Zune 30GB media player froze simultaneously. The device driver converted an internal day-count into a calendar year using a loop resembling:

```
while days > 365:
    if is_leap_year(year):
        if days > 366:
            days -= 366
            year += 1
        else:
            days -= 365
            year += 1
```

On day 366 of a leap year, the outer condition (`days > 365`) was still true, but the inner condition (`days > 366`) was false — so nothing inside the loop ever executed, `days` never changed, and the loop ran forever. The code correctly handled `days > 366`, but not the boundary case `days == 366`. This is a textbook boundary-value defect: the fault was invisible for 365 days out of every non-leap year and 365 out of 366 days in a leap year, surfacing only on one specific boundary date. The devices resolved themselves automatically at midnight, once the day count advanced past the boundary.

5 Negative Testing

5.1 Positive vs. Negative Tests

A **positive test** checks that the software does the right thing with *valid* input[3] — the “happy path.” A **negative test** checks that the software does the right thing with *invalid* input:[3] input that is out of range, the wrong type, missing, malformed, or otherwise not what the specification expects.

Key Idea

“The right thing” for invalid input is almost never “crash with an unhandled, confusing error,” and it is even worse to silently produce a wrong-but-plausible-looking result. The right thing is usually a well-defined, well-documented failure: a clear exception with an informative message, a validation error returned to the caller, or a logged rejection — something a calling program (or a human) can detect and handle deliberately.

5.2 What Counts as Invalid Input?

Negative test cases typically fall into a few recognizable categories:

- **Out-of-range values:** a negative age, a percentage above 100, a date in the far future.
- **Wrong type:** passing a string where a number is expected, or `None` where an object is expected.
- **Missing or empty data:** an empty string, an empty list, a missing required field.
- **Malformed data:** an email address with no @, a date string in the wrong format, invalid JSON.
- **Extreme or adversarial values:** an extremely long string, unusual Unicode characters, unexpected whitespace.

Notice that the invalid equivalence classes from Section 3.2 (out-of-range values) are one important *source* of negative test cases, but negative testing is broader: it also includes invalid values that do not even fit neatly into the same equivalence classes as the valid input, such as wrong data types entirely.

5.3 Negative Testing Exposes What EP Alone Misses

The `calculate_late_fee()` example returns throughout this course precisely because it illustrates a negative-testing gap so clearly. The original equivalence classes considered for `days_late` were implicitly “typical small numbers” and “numbers large enough to trigger the cap.” Nobody defined an invalid class for negative values, because the requirement never said what should happen if `days_late` were negative — so no test, positive or boundary, was ever written for it:

```
>>> calculate_late_fee(-4)
-6.0
```

A systematic negative-testing pass asks explicitly: *what happens if this parameter is negative? Zero? Not a number at all?* — for every parameter, regardless of whether the specification says anything about it. This is exactly the discipline that would have caught this fault before release, and it is why negative testing is not optional polish; it is one of the most cost-effective testing activities you can perform, precisely because specifications so often say nothing about invalid input.

Real-World Case Study

The Year 2000 Problem (Y2K).^[8, 9] For decades, many software systems stored calendar years using only two digits ("99" instead of "1999") to save memory, which was expensive when the practice began. This design implicitly treated “two-digit year” as the *only* equivalence class worth handling — no one systematically asked what should happen at the rollover boundary from "99" to "00". When date comparisons or arithmetic assumed the year always increased, a rollover to "00" could be interpreted as 1900 instead of 2000, corrupting age calculations, sort order, and interest calculations. Fixing potentially affected systems ahead of the January 1, 2000 rollover became one of the largest coordinated software remediation efforts in history, with global cost estimates commonly cited in the \$300–600 billion range (a wide, disputed range, since no single audited total exists). Y2K is a boundary-value story at civilization scale: a boundary condition (the century rollover) that was foreseeable decades in advance, and expensive precisely because it was not designed for early.

6 Putting It Together: A Worked Example

Consider a shipping-cost function with the following specification:

Given a package weight in kilograms, return the shipping cost: \$5.00 for $0 < \text{weight} \leq 1$ kg, \$10.00 for $1 < \text{weight} \leq 5$ kg, and \$20.00 for $5 < \text{weight} \leq 20$ kg. Weights of 0 kg or less, or above 20 kg, are invalid and should raise a `ValueError`.

Here is one plausible implementation:

```
def determine_shipping_cost(weight_kg):
    if weight_kg <= 0:
        raise ValueError("weight_kg must be positive")
    if weight_kg > 20:
        raise ValueError("packages over 20kg require freight shipping")
    if weight_kg <= 1:
        return 5.00
    elif weight_kg < 5:      # <-- suspicious: compare to the spec above
        return 10.00
    else:
        return 20.00
```

6.1 Step 1: Equivalence Classes

Class	Description	Valid?	Example
EC1	$0 < \text{weight} \leq 1$	Valid	0.5
EC2	$1 < \text{weight} \leq 5$	Valid	3.0
EC3	$5 < \text{weight} \leq 20$	Valid	10.0
EC4	$\text{weight} \leq 0$	Invalid	-1.0
EC5	$\text{weight} > 20$	Invalid	25.0

6.2 Step 2: Boundary Values

There are four boundaries in the specification: 0, 1, 5, and 20.

Boundary	Just Below	At Boundary	Just Above	Expected
0	-0.01 (invalid)	0.0 (invalid)	0.01 (EC1, \$5)	see below
1	0.99 (EC1, \$5)	1.0 (EC1, \$5)	1.01 (EC2, \$10)	see below
5	4.99 (EC2, \$10)	5.0 (EC2, \$10)	5.01 (EC3, \$20)	see below
20	19.99 (EC3, \$20)	20.0 (EC3, \$20)	20.01 (invalid)	see below

Look closely at the boundary of 5: per the specification, $\text{weight} = 5.0$ belongs to EC2 ($1 < \text{weight} \leq 5$) and should cost \$10.00. But the implementation's `elif weight_kg < 5` is a **strict** less-than — so `weight_kg = 5.0` falls through to the `else` branch and incorrectly returns \$20.00. This is precisely the class of fault BVA exists to catch, and precisely why testing only an interior representative like 3.0 for EC2 (as equivalence partitioning alone might suggest) would never reveal it.

6.3 Step 3: Negative Test Cases

Beyond the invalid numeric ranges already captured in EC4 and EC5, a thorough negative-testing pass also asks what happens with invalid *types*:

```
>>> determine_shipping_cost(None)
Traceback (most recent call last):
...
TypeError: '<=' not supported between instances of 'NoneType' and 'int'
```

The function does fail — but with a confusing, implementation-level error message that leaks Python internals to the caller instead of a clear, documented `TypeError` or `ValueError` explaining what went wrong. This is a real (if minor) defect that only a deliberate negative test for “wrong type entirely” would surface; it lives outside any equivalence class defined purely in terms of numeric ranges.

6.4 Step 4: Encode the Design as a pytest Suite

The three techniques above combine naturally into one parametrized `pytest` suite, exactly as in the *pytest* reading:

```
import pytest

# Equivalence partitioning: one representative value per valid class,
# plus boundary-value analysis: the exact edges of every class.
@pytest.mark.parametrize(
    "weight_kg, expected_cost",
    [
        # EC1: 0 < weight <= 1
        (0.01, 5.00),
```

```

        (0.5, 5.00),
        (1.0, 5.00),
        # EC2: 1 < weight <= 5 (boundary at 5.0 currently FAILS -- see bug)
        (1.01, 10.00),
        (3.0, 10.00),
        pytest.param(5.0, 10.00, marks=pytest.mark.xfail(reason="boundary bug: < should
be <=")),
        # EC3: 5 < weight <= 20
        (5.01, 20.00),
        (10.0, 20.00),
        (20.0, 20.00),
    ],
)
def test_valid_weights(weight_kg, expected_cost):
    assert determine_shipping_cost(weight_kg) == expected_cost

# Negative testing: invalid numeric ranges (EC4, EC5) and the boundaries
# immediately outside the valid region.
@pytest.mark.parametrize("weight_kg", [-1.0, -0.01, 0.0, 20.01, 25.0])
def test_invalid_weight_raises_value_error(weight_kg):
    with pytest.raises(ValueError):
        determine_shipping_cost(weight_kg)

# Negative testing: invalid types entirely.
@pytest.mark.parametrize("bad_input", [None, "5", [5], {}])
def test_invalid_type_raises_a_clear_error(bad_input):
    with pytest.raises((TypeError, ValueError)):
        determine_shipping_cost(bad_input)

```

Running this suite reports the known boundary bug as an expected failure (XFAIL) rather than silently passing or breaking the build, exactly as in the *pytest* reading — and every other case, spanning three equivalence classes, four boundaries, and two flavors of negative input, passes with a total of eighteen focused test cases instead of an arbitrary, unsystematic handful.

7 How Much Is Enough?

Equivalence partitioning and boundary-value analysis do not tell you to test *every* boundary of *every* parameter of *every* function in a codebase with equal intensity. Recall Principle 6 from the *Testing Foundations* reading: **testing is context-dependent**. A boundary in a billing calculation used by every customer deserves careful BVA; a boundary in an internal debug-logging helper may not.

Tip

A practical rule of thumb: invest the most in equivalence partitioning and boundary-value analysis for code that (a) implements business rules with explicit numeric or categorical thresholds, (b) handles money, safety, or security, or (c) has a history of off-by-one defects. These are exactly the places where a boundary mistake is both likely and costly.

8 Common Pitfalls

Common Pitfall

Testing only the “obvious middle” of a class. Choosing 50 as “a representative valid percentage” feels natural, but it is the least likely value to expose an off-by-one defect. Always pair equivalence partitioning with boundary-value analysis rather than using EP alone.

Common Pitfall

Defining invalid classes only for what the spec explicitly forbids. A specification that says “accepts an integer age” rarely spells out what should happen for a negative age, a non-integer, or `None` — but all three are inputs a real caller can and eventually will pass. Ask about every parameter, not just the ones the specification happens to mention.

Common Pitfall

Confusing “the function didn’t crash” with “the function handled it correctly.” A negative test should assert on *what* happens for invalid input (a specific exception type, a specific error message, a specific return value) — not merely that the program kept running.

9 A Practical Mental Model

When you see a specification like “accepts values from 1 to 100,” think:

“Three classes: below 1, from 1 to 100, above 100. Six boundary values: 0, 1, 2 and 99, 100, 101.”

When you see a comparison operator in code (`<`, `<=`, `>`, `>=`), think:

“What is the exact input that makes this condition flip? That is my next test case.”

When you finish writing “happy path” tests for a function, think:

“What is every way a caller could hand this function garbage — wrong type, missing, empty, out of range — and does the function fail clearly when they do?”

10 Summary

Term	Meaning
Equivalence class	A set of inputs the software is expected to handle identically
Equivalence partitioning (EP)	Testing one representative value per equivalence class
Valid / invalid class	Inputs the spec accepts / inputs the spec should reject
Boundary-value analysis (BVA)	Testing at, just below, and just above every class boundary
Positive test	A test using valid input (the “happy path”)
Negative test	A test using invalid input, checking for graceful, well-defined failure

Key Idea

Test the edges, and test the garbage. Equivalence partitioning gives you a small, representative set of inputs instead of an infeasible exhaustive one; boundary-value analysis focuses your limited testing budget exactly where defects are statistically most likely to hide; negative testing makes sure your software fails safely and clearly when it inevitably receives input the specification never anticipated.

11 Practice Exercises

Exercise 1 — Equivalence Classes

A function `classify_bmi(bmi)` returns "underweight" for $\text{BMI} < 18.5$, "normal" for $18.5 \leq \text{BMI} < 25$, "overweight" for $25 \leq \text{BMI} < 30$, and "obese" for $\text{BMI} \geq 30$. Identify all equivalence classes (there is no invalid numeric range specified here — but consider what *type*-related invalid classes might still exist).

Exercise 2 — Boundary Values

For the `classify_bmi()` function above, list every boundary and, for each one, the three boundary-value test inputs (just below, at, just above).

Exercise 3 — Finding the Bug

Using only the boundary values you listed in Exercise 2, determine whether this implementation is correct, and if not, identify the exact input that reveals the fault:

```
def classify_bmi(bmi):
    if bmi < 18.5:
        return "underweight"
    elif bmi <= 25:
        return "normal"
    elif bmi < 30:
        return "overweight"
    else:
        return "obese"
```

Exercise 4 — Negative Testing

Write a list of at least six negative test cases for a function `parse_percentage(text)` that is supposed to parse a string like "42%" into the float 0.42. Include at least one case each for: wrong type, malformed format, out-of-range numeric value, and empty input.

Exercise 5 — Design Question

A teammate says: “We don’t need boundary-value analysis if we already have equivalence partitioning — picking one value per class is enough.” Using the `determine_shipping_cost()` example from this reading, write a short argument for why this claim is incorrect.

12 Discussion Questions

1. Why does equivalence partitioning rely on an assumption that might not always be true? What is that assumption?
2. Why do defects cluster at boundaries more than in the interior of an equivalence class?

3. How does boundary-value analysis relate to the specific comparison operators ($<$, $<=$, $>$, $>=$) used in an implementation?
4. Why is it not enough to test only the invalid ranges the specification explicitly mentions?
5. What does it mean for a negative test to check for “graceful” failure, and why does that matter more than simply confirming the program didn’t crash?
6. How does negative testing relate to the “absence-of-errors fallacy” principle from the *Testing Foundations* reading?
7. Why might a team choose not to apply full combinatorial testing across every parameter of a multi-parameter function?
8. In the Zune case study, why did the bug depend on the *specific date* rather than being present at all times?
9. How is the Y2K problem an example of an unexamined equivalence class (“years always increase”) rather than a single line-of-code bug?
10. How would you decide how much boundary-value and negative testing is “enough” for a given function in a real project?

13 Key Takeaways for SE 413/513

1. **Systematic beats exhaustive.** Equivalence partitioning and boundary-value analysis exist because exhaustive testing is infeasible, not merely inconvenient.
2. **Partition first, then target the edges.** Use equivalence partitioning to identify classes; use boundary-value analysis to choose the highest-value test inputs within and around them.
3. **Every parameter has invalid classes, whether the spec says so or not.** Negative values, wrong types, `None`, and empty input are always worth asking about.
4. **A comparison operator is a promise to test.** Every $<$, $<=$, $>$, or $>=$ in a specification or implementation names an exact input worth a dedicated test case.
5. **Negative testing checks for graceful failure, not just survival.** Assert on the specific exception or error behavior, not merely that nothing crashed.
6. **These techniques are context-dependent.** Invest more testing effort in code that handles money, safety, security, or business-critical thresholds.
7. **Small, well-chosen test suites beat large, unsystematic ones.** A dozen carefully derived equivalence-class and boundary tests typically find more real defects than fifty arbitrarily chosen ones.

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